

Configuring Basic Network Hardware

Task 1: Router Admin Login

Procedure

1. Connected the laptop to the home Wi-Fi network.
 2. Opened **Command Prompt**.
 3. Typed **ipconfig** and pressed **Enter**.
 4. Noted the **Default Gateway** IP address.
 5. Opened a web browser and entered the gateway address (for example, **192.168.1.1**).
 6. The router login page opened successfully.
-

Task 2: Current Network Configuration

Network Setting Example Value

IP Address 192.168.1.105

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

Task 3: Configure a Static IP Address

Example Static IP Settings

Setting	Value
IP Address	192.168.1.50
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
Preferred DNS	8.8.8.8
Alternate DNS	8.8.4.4

Steps

1. Open **Network & Internet Settings**.
2. Open **Wi-Fi** (or **Ethernet**) properties.
3. Change **IP Assignment** from **Automatic (DHCP)** to **Manual**.
4. Enter the static IP configuration shown above.
5. Save the settings.
6. Verify that the internet connection still works.
7. Change the setting back to **Automatic (DHCP)** after testing.

Observation

The device connected successfully using the static IP address and internet access was available. The settings were then reverted to DHCP to avoid future IP conflicts.

Task 4: DHCP Range

Example DHCP Configuration

Setting	Example Value
Router IP	192.168.1.1
DHCP Start	192.168.1.100
DHCP End	192.168.1.199

Explanation

The DHCP server automatically assigns unique IP addresses to devices connected to the network. If two devices are assigned the same IP address, an **IP address conflict** occurs. This can cause network interruptions, loss of internet connectivity, and communication problems because the router cannot correctly identify which device should receive the network traffic.

Conclusion

The router was accessed using its default gateway address, the device's network configuration was identified, a static IP address was successfully configured and tested, and the DHCP range was examined. This practical demonstrated how IP addressing and DHCP help manage communication within a network.